Independent audit: defect spectrum, fusion assumptions, and the Toda frame

Quantum Boussinesq solitons project

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Abstract

The proposed Toda expansion can proceed independently, but the quantum identification still needs work. **(d)** An exact symbolic calculation finds a factor of sixteen missing from the displayed defect sewing determinant. Its zeros are unaffected, but the stated growing-mode formula also misses a change of admissible branch. **(c)** The earlier warning that a KP triangle passes through unstable defect vacua remains relevant to the claimed quantum fusion selection rule. The edge-reference subtraction, being independent of the second soliton, cannot by itself remove the linear ratio dependence attributed to it in Phase 2h. **(c)**

1 Scope and reproducibility

This note supplements `independent_phase2h_audit.tex` and the separate charge and determinant audits. Labels (a)–(d) retain their repository meaning; (c) denotes a project derivation or check and (d) an interpretation or research judgment. No external literature claim is introduced here. We retain the good-equation and Corrigan–Zambon dictionaries in `AGENTS.md`; the sewing matrix below uses the existing CZ field normalisation. The audit does not choose a regularisation or quantum phase convention.

All displayed new mathematical results are checked in `phase2/scripts/audit_frames_defects.py`, with exact SymPy identities and 70-digit mpmath calculations. Its JSON companion records numerical results and SHA-256 hashes of the audited inputs. The previously existing `oneloop_defect_scattering_and_stability.py` also completes with `ALL CHECKS PASSED`; its tests do not expose the issues below.

2 The defect determinant: check away from its zeros

Corrected normalisation (c) . Let $s = \eta\sigma$, and let $\omega = ig$ with $g > 0$ be a purely growing frequency. The decaying half-line spatial exponents satisfy

$$\kappa_1^2 + \kappa_2^2 = 1, \qquad \kappa_1\kappa_2 = g, \qquad S = \kappa_1 + \kappa_2 = \sqrt{1+2g}, \qquad d = \kappa_1 - \kappa_2, \quad d^2 = 2 - S^2.$$

For the actual 4×4 linearised sewing matrix implemented in Phase 2g, at $\eta = +1$, the exact factorisation is

$$\frac{\det M}{(\kappa_1 - \kappa_2)^2} = -\frac{2}{3} S^2 (S^2 - 1) P_-(S; s) P_+(S; s), \tag{1}$$

where the reported polynomials themselves are correct:

$$P_\mp(S; s) = S^2 - \frac{\sqrt{3} \mp 1}{2} sS + \frac{s^2 - 2 \mp \sqrt{3}}{2}.$$

The coefficient in Phase 2g, Eq. (`det_factorisation`), is $-1/24$, sixteen times too small. The script proves (1) symbolically by replacing each half-line pair of columns by its sum and its

difference divided by d, expanding the determinant, and imposing $d^2 = 2 - S^2$. It also compares an independently evaluated determinant to the existing matrix at six nonzero samples: $s = -1.5, 0.5, 1.9$, $g = 0.2, 0.7$. The ratio of actual to displayed determinant is sixteen in every case.

This constant does not change the eigenvalue zeros, and it is not by itself a computed error in the physical one-loop determinant. **(d)** The existing factorisation test evaluates the two sides at a characteristic root, where both vanish. It therefore does not test the prefactor. **(c)**

Admissible roots (c) . For $-2 < s < 2$, retain only real roots with $S > 1$ from

$$S_{-,\pm} = \frac{(\sqrt{3}-1)s \pm (\sqrt{3}+1)\sqrt{4-s^2}}{4},$$
$$S_{+,\pm} = \frac{(\sqrt{3}+1)s \pm (\sqrt{3}-1)\sqrt{4-s^2}}{4}, \qquad g = \frac{S^2-1}{2}.$$

Squaring an inadmissible root loses the decaying-sheet condition. The two positive growth rates at $s = 1.9$ are approximately

$$g = 0.200256\ldots, \qquad 0.496894\ldots;$$

the JSON gives all digits. The smaller mode comes from $S_{+,-} > 1$; $S_{-,+} < 1$ has ceased to describe a growing mode. The displayed frequency pair in Phase 2g instead includes the negative of that smaller rate. Direct substitution of both admissible rates into the sewing determinant gives zero to the 70-digit calculation's tolerance.

At $s = 2$ one has

$$P_+(S;2) = \left(S - \frac{\sqrt{3}+1}{2}\right)^2, \qquad g = \frac{\sqrt{3}}{4}.$$

Thus the admissible roots coalesce at the same positive growth rate. The displayed frequency pair gives opposite signs there, contradicting the coalescence stated immediately afterwards. The existing script tests $s = 1.5$ and then uses a separate expression at $s = 2.5$; it skips the intervening branch change. These are explicit counterexamples to that formula as a complete spectrum, not a replacement completeness proof.

3 Fusion: an algebraic identity does not supply missing states

Stable-vertex check (c) . The KP triple may be parametrised as

$$q_j = \frac{1}{\sqrt{3}} \cos\left(\vartheta + \frac{2\pi j}{3}\right), \qquad \sigma_j = -2\sqrt{3} q_j, \qquad j = 0, 1, 2.$$

The script verifies the pair relation $q_i^2 + q_i q_j + q_j^2 = 1/4$ and counts admissible growing modes on 198 oriented, nondegenerate sampled triples. Each has exactly one vertex with no growing mode, agreeing with the previously established condition $\eta\sigma < -1$. This is a sample check, not a new proof of spectral completeness at thresholds.

`defect_semiclassics.tex`, its stability section, explicitly downgraded the proposed exact quantum triangle bootstrap because its two excited defect vacua are unstable. Phase 2g proves a continuum phase coboundary identity and then calls it a decisive quantum selection rule; that step does not restore stable intermediate states. **(d)** Nor does a continuum identity determine the contribution of any missing discrete modes or the normalisation of the quantum defect transitions. An analytic continuation involving resonances might still obey useful identities, but that needs its own derivation. We retain the algebraic fusion result and do not treat it alone as a physical exclusion of a Gamma factor or a selection of an exchange-phase constant. **(d)**

4 The edge subtraction and the quantum frame

What the code subtracts (c) . In `toda_edge_twoterm.py`, the phase numerator is

$$\Phi(p_1, p_2) - \Phi(p_1, e_2) - (-2\kappa_1)\left(\frac{P_{n2}}{a} - \Pi_2\right).$$

The frame term uses the classical quantities $P_{n2} = 4b(\kappa_2 \cosh\kappa_2 - \sinh\kappa_2)$ and $\Pi_2 = 2b\sinh\kappa_2$ in the right-moving sector. With $\kappa_j = \sqrt{y_j\hbar_T/b}$, the script verifies

$$\lim_{\hbar_T\to 0} \frac{(-2\kappa_1)(P_{n2}/a - \Pi_2)}{\hbar_T} = 4\sqrt{y_1 y_2}.$$

This establishes the implemented classical subtraction. It does not establish that the same transformation has no quantum correction. `toda_edge.tex` explicitly retains that caveat, as does `matching_toda.py`. A complete comparison to continuum quantum scattering still needs a common quantum definition of the frame and of the physical charge labels. **(d)**

An edge value cannot subtract a slope (c) . Phase 2h attributes the $+4r/\pi$ term in its candidate limiting residual to subtracting the edge reference $-\Phi(p_1, e_2)/\hbar_T$, saying that this removes the linear small-r slope. At fixed $y_1 = Y$ the ground state, p_1, and e_2 are independent of $y_2 = r^2 Y$. Therefore this subtraction is a function $C(Y)$ alone, and exactly

$$\partial_r\{F(Y, r) - C(Y)\} = \partial_r F(Y, r).$$

Equivalently, differences between any two ratios are unchanged. It cannot alone add the non-constant function $4r/\pi$. This invalidates that explanation of the term, without disproving the candidate final residual: the term might instead come from the dressed phase, charge conversion, or another part of a derivation not yet supplied. **(d)** The earlier audit already records that the script defines the purported loop asymptotic by subtracting the classical contribution from the proposed target.

5 Checks, failures recorded, and verdict

Claim or scope	Verification
Corrected sewing determinant, away from zeros (c)	Exact polynomial identity; six 70-digit samples
Growing-mode counterexample and coalescence (c)	Root filtering, direct determinants, exact double root
One stable vertex on sampled triples (c)	198 oriented nondegenerate triples
Classical frame limit; edge subtraction preserves slope (c)	Exact SymPy identities
Physical quantum frame and fusion selection (d)	Still require derivation

Failures recorded.

1. The displayed factorisation fails away from its roots by a factor of sixteen. The stated frequency pair fails the admissible growing-branch test at $s = 1.9$ and the stated coalescence at $s = 2$.

2. An exploratory call to mpmath's LU determinant raised a `TypeError` at an exactly singular matrix. The independent checker uses the exact 24-term determinant definition, retaining the same matrix and tolerance; no eigenvalue condition is dropped.

3. A direct symbolic determinant computation was interrupted after stalling. Symmetric and antisymmetric column combinations reduce its size and prove the same off-shell identity exactly.